# A Decision Support System for Agricultural Profit-Sharing: Design, Risk Analysis, and Performance Evaluation

## UNDERGRADUATE THESIS


**Submitted by**

Hani Ayu Fadila (4337855201230026)
Liana Syifa Fauzia (4337855201230112)
Ririn Dwi Ariyanti (4337855201230045)

**INFORMATICS STUDY PROGRAM**
**FACULTY OF INFORMATION TECHNOLOGY AND COMPUTER**
**HORIZON INDONESIA UNIVERSITY**


**2027**

## STATEMENT OF ORIGINALITY

We, the undersigned, hereby declare that the undergraduate thesis entitled:

**A Decision Support System for Agricultural Profit-Sharing: Design, Risk Analysis, and Performance Evaluation**

is entirely our own work and is not a duplication of any other party's work. This thesis report does not contain any scientific work or opinions that have previously been written or published by others, except for those that are properly cited in the text and listed in the references.
If it is later proven that this thesis contains elements of plagiarism, we are willing to accept that this thesis be declared invalid, that any academic degree obtained be revoked, and that we be processed in accordance with applicable laws and regulations.

This statement is made truthfully and in full awareness of the consequences.

Karawang, February 26, 2027

| | | |
|---|---|---|
| *(Materai 10.000)*<br>*(Tanda tangan)* | *(Materai 10.000)*<br>*(Tanda tangan)* | *(Materai 10.000)*<br>*(Tanda tangan)* |
| Hani Ayu Fadila | Liana Syifa Fauzia | Ririn Dwi Ariyanti |

# APPROVAL PAGE

THESIS TITLE

## A Decision Support System for Agricultural Profit-Sharing: Design, Risk Analysis, and Performance Evaluation

Submitted by

| | | | | |
|---|---|---|---|---|
| **Name** | : | Hani Ayu Fadila | Liana Syifa Fauzia | Ririn Dwi Ariyanti |
| **NIM** | : | 4337855201230026 | 4337855201230112 | 4337855201230045 |
| **Study Program** | : | Informatics | Informatics | Informatics |

Approved by:

**Thesis Supervisor 1**

*(Signature)*

**Name of Supervisor 1**
NIDN.

**Thesis Supervisor 2**

*(Signature)*

**Name of Supervisor 2**
NIDN.

**Head of Informatics Study Program**
**(Nama program studi)**

*(Signature)*

**Name Kaprodi**
NIDN.

**Dean FTIK**

*(Signature)*

**Name Dekan**
NIDN.

# ENDORSEMENT PAGE

THESIS TITLE

## A Decision Support System for Agricultural Profit-Sharing: Design, Risk Analysis, and Performance Evaluation

Submitted by

| | | | | |
|---|---|---|---|---|
| **Name** | : | Hani Ayu Fadila | Liana Syifa Fauzia | Ririn Dwi Ariyanti |
| **NIM** | : | 4337855201230026 | 4337855201230112 | 4337855201230045 |
| **Study Program** | : | Informatics | Informatics | Informatics |

**Declared to have passed the Thesis Examination after successfully defending the thesis before the Board of Examiners of the Study Program Informatics Engineering**

Karawang, February 26, 2027

Board of Thesis Examiners

**Examiner 1**

*(tanda tangan)*

**Nama Penguji 1**
NIDN.

**Examiner 2**

*(tanda tangan)*

**Nama Penguji 2**
NIDN.

# ABSTRACT

The Indonesian agricultural sector faces major challenges, especially from the side of farmers and investors. Where farmers who own land do not have the capital to manage their land optimally, while investors who have capital are hesitant to invest with farmers, because of uncertainty regarding the expected results will not be in accordance with what is expected. Many cases where investors actually suffer large losses from investments, so a digital tool is needed that can help the risk of crop failure for investors and help farmers get capital to manage their land, without having to accept losses from both parties. This study aims to design and develop a mobile and web-based agricultural platform using the Design Thinking method, which is equipped with a Decision Support System feature for profit sharing with an AI Risk Estimator as its main feature. The system uses a rule-based artificial intelligence approach to predict the risk of crop failure based on data on crop types, land location, agricultural history, and weather conditions. This platform is expected to help investor farmers transparently through a contract system for digital profit sharing with real-time performance tracking. System testing was carried out using Black Box Testing and System Usability Scale (SUS) to evaluate functionality and ease of use. This research is expected to provide real solutions for farmers' access to capital while providing a safer and more transparent agricultural investment experience for investors.



***Keyword: Decision Support System, Profit-Sharing, AI Risk Estimator, Agricultural Platform***


---

# PREFACE

First and foremost, praise and gratitude are extended to Allah SWT for His blessings and grace, which have enabled the author to complete this undergraduate thesis report. This report is submitted as one of the requirements for obtaining a Bachelor's degree (S1) in the Informatics Study Program, Faculty of Information Technology and Computer Science, Horizon Indonesia University.

The author realizes that this thesis report is still far from perfect. Therefore, constructive criticism and suggestions are highly appreciated. Despite all limitations, the author also acknowledges that this thesis report would not have been possible without the support, guidance, and encouragement from various parties. Thus, with sincere gratitude, the author would like to express appreciation to:

1. …… as the Rector of Horizon Indonesia University;
2. …… as the Dean of the Faculty of Information Technology and Computer Science:
3. …… as the Head of the Study Program …… *(sesuaikan dengan program studi penulis)*
4. …… as the First Thesis Advisor;
5. …… as the Second Thesis Advisor;
6. The author’s beloved parents *(bisa disesuaikan penulisannya)*
7. …… *(selanjutnya dapat disesuaikan dengan kebutuhan penulis)*

May Allah SWT reward all kindness and always bestow His guidance, blessings, and grace upon them. Aamiin.

Karawang, January 2, 2025
*(disesuaikan dengan tanggal dikumpulnya laporan skripsi)*

(tanda tangan penulis)

Anwar Hilman
*(nama penulis)*

# Table of Contents

**ABSTRACT** **5**
**PREFACE** **6**
**Table of Contents** **7**
**CHAPTER 1**
**INTRODUCTION** **8**
1.1 Background 8
1.2 Problem Statement 9
1.3 Research Objectives 9
1.4 Research Benefits 10
1.5 Problem Scope 11
1.6 Research Methodology 11
1.7 Report Structure 12
**Chapter 2: Literature Review** **13**
2.1 Key Definitions and Concepts 13
2.2 Related Work 13
2.3 Technologies Used 13
2.4 Theoretical Framework 13
**Chapter 3: System Design and Implementation** **14**
**Chapter 4: Results and Discussion** **15**
**Chapter 5: Conclusions and Recommendations** **16**
**References** **16**
**Appendices** **18**

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

The agricultural sector is widely recognized as one of the most important sectors in Indonesia's economy. However, despite its importance, it still faces a number of ongoing structural problems. One issue that often occurs is the gap between farmers and investors. Many farmers already have access to land, but they do not have enough capital to manage it properly. Meanwhile, investors who actually have the financial resources tend to be more careful when it comes to investing in agriculture. This mainly happens because agriculture involves a high level of uncertainty, particularly due to the risk of crop failure.

As a result, much of the available agricultural land is not utilized to its full potential. In addition, many investment opportunities that could have been beneficial are not taken advantage of. In the long run, this condition affects not only agricultural output but also the economic development of rural areas. There have been several digital solutions developed to respond to this issue. However, most of them still focus on certain parts of the problem and have not completely addressed the gap between farmers and investors.SIPROTANI (Dewi et al., 2022) developed a mobile-based agricultural information system focused on documenting and reporting agricultural products using the Agile/Scrum method, yet it does not address financial transactions, investment mechanisms, or risk assessment between farmers and investors. Similarly, the Petani Cerdas application (Wijaya et al., 2024) provides agricultural training and consultation features for farmers using an evolutionary prototype method, but lacks any investment system or AI-based risk estimation capability. A broader study by Prayoga (UGM) reviewing various agricultural digital platforms in Indonesia, including iGrow and TaniHub, found that while some platforms offer profit-sharing investment concepts, none integrate an automated AI risk estimator accessible directly by investors. Furthermore, research by Palupi et al. (2021) analyzing the fintech platform Crowde revealed that crop failure risk, referred to as Non-Performing Loan (NPL), remains a critical challenge, yet risk assessment is still conducted manually by the company's operational team rather than through an automated digital system. Most recently, Khuluq and Sriyono (2025) emphasized the strategic importance of integrating financial technology with digital agricultural platforms, noting that most existing studies address each domain separately without providing an integrated system featuring intelligent risk assessment.

This research therefore aims to design and develop a mobile and web-based agricultural platform using the Design Thinking methodology, integrating a Decision Support System with an AI Risk Estimator as its core feature. The system utilizes a rule-based artificial intelligence approach to predict crop failure risks based on plant type, land location, agricultural history, and weather conditions, while also facilitating a transparent digital profit-sharing contract system between farmers and investors with real-time performance tracking. This solution is expected to bridge the existing gap in agricultural technology by providing a safer, more transparent, and data-driven investment ecosystem that empowers both farmers and investors in Indonesia.

### 1.2 Problem Statement

Based on the background described above, the research problem can be defined as follows: How can a mobile and web-based agricultural platform be developed using the Design Thinking methodology that integrates a Decision Support System with an AI Risk Estimator to predict crop failure risks and support a transparent digital profit-sharing system between farmers and investors?

### 1.3 Research Objectives

Based on the problem identification previously established, the objectives of this research are:

1. To design and develop a mobile and web-based agricultural platform using the Design Thinking methodology that integrates a transparent digital profit-sharing system between farmers and investors.
2. To integrate a Decision Support System (DSS) featuring a rule-based AI Risk Estimator capable of predicting crop failure risks based on crop type, land location, agricultural history, and weather conditions.
3. To construct a real-time agricultural performance tracking system to provide accurate data for investors in making informed investment decisions.
4. To validate the effectiveness of an intelligent system in mitigating investment risks (reducing Non-Performing Loans) while optimizing the utilization of agricultural land in Indonesia.

### 1.4 Research Benefits

This study is built to close the gap between farmers and investors through smart technology. We've broken down the benefits into two categories:

**1. Academic Contributions**
This research adds fresh perspectives to Agricultural IT and Decision Support Systems by:

a. Offering a transparent scholarly reference regarding how the Design Thinking model can address complex issues involving multiple participants in agriculture.
b. Assisting in the development of more intelligent agricultural failure predictions by employing artificial intelligence to evaluate past farming practices, data unique to certain locations, and climatic patterns.
c. Delivering a robust scientific method for evaluating digital platforms, guaranteeing their seamless functionality and delivering an excellent experience for the user.

**2. Practical Contributions**
On the ground, this research provides tools that make a real difference in the digital farming world:

a. Farmers

It paves the way for smoother financial support by eliminating excessive bureaucracy. With enhanced availability of funds, agriculturalists can concentrate on optimizing their land and utilizing superior resources to boost their production.

b. Investors
The AI Risk Estimator removes uncertainty from the investment process. It provides investors with clear, data-based information prior to making any financial commitments, assisting them in avoiding poor transactions and reducing possible losses.

c. Farming Ecosystem
With real-time performance tracking, the entire process becomes transparent. This builds genuine trust between investors and farmers, cuts out communication gaps, and makes the whole food supply chain run much more effectively.

### 1.5 Problem Scope

This study focuses on the development of a Decision Support System (DSS) for profit-sharing schemes in the agricultural sector, specifically to assist investors in assessing investment risks and to help farmers communicate information regarding agricultural conditions.

This research addresses several key issues, including:

1. a digital system connecting farmers and investors to enhance information transparency between them within the profit-sharing scheme
2. an AI Risk Assessment feature to estimate the potential for crop failure, which currently makes it difficult for investors to assess this risk before investing
3. An information dashboard to support investment decisions, as there is currently no system capable of providing risk estimates as a basis for decision-making

The limitations not covered in this research include:

1. Does not address system implementation at a production scale or real-world field applications
2. Does not cover direct financial transactions
3. Does not address the physical distribution of harvests or logistics
4. Does not use complex machine learning-based AI models (only rule-based)
5. Additional features such as marketplaces and donations are not the primary focus of this research

The prototype to be developed in this research consists of:

1. A web application used by investors to:
    a. View agricultural project information
    b. View risk analysis results
    c. Support the decision-making process

2. A mobile application used by farmers to:

    a. Input agricultural data
    b. Update the condition and progress of crop fields

### 1.6 Research Methodology

This study employs the design thinking method, an approach that focuses on user needs when designing solutions to a problem. The stages of the Design Thinking method used are as follows:

1. **Empathize**

   This stage aims to understand the needs and challenges of the users, namely farmers and investors. Activities conducted in this stage include a literature review on agricultural profit-sharing systems and an analysis of issues such as the risk of crop failure and information gaps.

2. **Define**

   In this stage, the problem is formulated based on the results of the Empathize stage. The primary issues identified are a lack of transparency in information between farmers and investors, the absence of a system to assist investors in assessing investment risks, and the lack of a decision-support system within the agricultural profit-sharing scheme.

3. **Ideate**

   This stage involves developing solution ideas to address the problems defined in the Define stage. Proposed solutions include the development of a digital system to connect farmers and investors, the integration of AI-based risk assessment features for estimating crop failure risks, and the provision of a dashboard as a decision-making tool.

4. **Prototype**

   In this phase, the design and development of the system prototype are carried out. The developed prototype includes a web application for investors and a Flutter-based mobile application for farmers, with the prototype development covering user interface (UI/UX) design, system flow design, and the implementation of key features such as the profit-sharing system and risk analysis.

5. **Test**

   This phase is conducted to test the developed system to determine whether it operates in accordance with the research objectives, and the results of this phase are used as the basis for evaluating and improving the developed system. The testing methods used

include Black Box Testing to assess system functionality and the System Usability Scale (SUS) to measure usability.

### 1.7 Report Structure

This thesis contains five chapters that explain the research step by step.

**Chapter 1 (Introduction) :** starts with the background of the study, focusing on the issue of the gap between farmers and investors in the agricultural sector. It also covers the problem formulation, research objectives, benefits, scope of the study, and the Design Thinking approach used in this research.

**Chapter 2 (Literature Review) :** explains several important concepts such as Decision Support Systems, AI-based risk estimation, and digital platforms in agriculture. In addition, this chapter reviews related studies and previous research, as well as the theories and technologies that support the development of this system.

**Chapter 3 (System Design and Implementation describes) :** the overall system design and architecture of the platform, including the mobile application for farmers and the web application for investors. It also explains the development process, tools used, and testing procedures conducted during system development.

**Chapter 4 (Results and Discussion) :** explains the results of the system implementation, including its performance, the usability evaluation based on Black Box Testing and the System Usability Scale (SUS), as well as a comparison with similar existing systems.

**Chapter 5 (Conclusions and Recommendations) :** presents the main findings of the research in line with the objectives, and also provides several suggestions for future improvements and further development of the system.